Spectral bounds for graph parameters – Part II

Pawel Wocjan

wocjan@cs.ucf.edu

University of Central Florida

(This talk is based on joint work with Clive Elphick.)

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This talk is the second of a pair of consecutive talks by Clive Elphick and Pawel Wocjan on their joint work on spectral bounds for graph parameters, that lie between the clique number $\omega(G)$, and the chromatic number, $\chi(G)$. The second talk will provide more technical details.

Let $[c] = \{0, \ldots, c-1\}$ for some positive integer c and let I_d and 0_d denote the identity and zero matrices acting on $\mathbb{C}^d$. A quantum c-coloring of an undirected graph $G = (V, E)$ without loops and without multiple edges is a collection of orthogonal projectors

$$\{P_{v,k} : v \in V, k \in [c]\}$$

acting on $\mathbb{C}^d$ for some $d > 0$ such that

- for all vertices $v \in V$
$$\sum_{k\in[c]} P_{v,k} = I_d \qquad \text{completeness}$$
- for all edges $vw \in E$ and for all $k \in [c]$
$$P_{v,k}P_{w,k} = 0_d \qquad \text{orthogonality}$$

The quantum chromatic number, χ_q, is the smallest c for which the graph G admits a quantum c-coloring. The classical chromatic number is a special case when $d = 1$.

We will establish that the existence of a quantum c-coloring implies the existence of a unitary matrix U acting on $\mathbb{C}^n \otimes \mathbb{C}^d$ such that

$$\sum_{\ell=1}^{c-1} U^\ell (A \otimes I_d)(U^*)^\ell = -A \otimes I_d,$$

where A is the adjacency matrix of G and n its number of vertices.

We will show that applying the majorization result for maximal eigenvalues of Hermitian matrices X and Y, $\mu_{\max}(X)+\mu_{\max}(Y) \geq \mu_{\max}(X+Y)$, to the above matrix equality yields the Hoffman bound for the quantum chromatic number: $\chi_q(G) \geq 1 + \mu_{\max(A)}/|\mu_{\min}(A)|$.

Similarly, we will show that applying the rank inequality for positive semidefinite matrices X and Y with $X - Y$ positive semidefinite, $\text{rank}(X) \geq \text{rank}(Y)$, to the above matrix inequality yields the inertial lower bound on the quantum chromatic number: $\chi_q(G) \geq 1 + \max\{n^+(A)/n^-(A), n^-(A)/n^+(A)\}$, where $n^\pm(A)$ denote the number of positive and negative eigenvalues of A.